

Dear Editor,

Thank you very much for arranging to review our manuscript. We have addressed all the questions/suggestions raised/made by Referee #1 and Referee #2. A detailed, point-by-point response to the comments of both referees has been provided below (each referee's comments are highlighted in blue). In view of these changes, we trust the manuscript will now be judged suitable for publication.

Sincerely,
Author

I. RESPONSE TO QUESTIONS/COMMENTS OF REFEREE #1

1. The author discusses Fermi arcs on the boundary of topological semimetals ... I think that the present paper may deserve publication in some form.

We thank the referee for this positive overall assessment of the manuscript and for the constructive suggestions below, which we address in turn.

2. However, as it was mentioned above the derivation of relation between Eq. (10) and Eq. (12) is to be added.

We agree that this step is a one-line calculation for the untilted Weyl node of Sec. II.A [cf. the paragraph containing Eqs. (6)–(8)], but requires genuine care once $H_N(\mathbf{k})$ contains higher (quadratic, cubic, or higher) powers of $k_\perp$, as is the case for several of the multifold and multi-Weyl models – precisely because the inverse decay length κ entering the surface-state ansatz is generically complex: normalisability fixes only $\kappa_r \equiv \text{Re}(\kappa) > 0$, while $\kappa_i \equiv \text{Im}(\kappa)$ is an independent real unknown, on the same footing as E [cf. the parametrisation below Eq. (11)]. We have therefore added a self-contained, general derivation, valid for arbitrary finite polynomial order in $k_\perp$ and for genuinely complex κ , as a new paragraph following Eq. (10), which we break up into four steps below and reproduce as a single paragraph in the revised manuscript:

“This general boundary term can be obtained by applying a Lagrange/Wronskian identity for higher-order derivative operators [cf. Eq. (45)] to each monomial order present in $H_N(k_\perp, \mathbf{k}_\parallel)$ in turn, and summing — the bulk contributions cancel identically, leaving the pure boundary functional $\Xi[\psi, \phi]$ evaluated at $r_\perp = 0$, with the $r_\perp \rightarrow \infty$ contribution vanishing by the exponential decay of ψ, ϕ . Specialising to the actual candidate surface state, $\phi = \psi = \psi_0 e^{-\kappa r_\perp}$, and using that this same state solves the eigenvalue problem, $H_N(\lambda) \psi_0 = E \psi_0$ with $\lambda \equiv i\kappa$ [Eq. (13)], the boundary functional collapses – exactly, and order by order in the momentum expansion of H_N , with no assumption that κ is real – onto $\Xi[\psi_0, \psi_0]|_{r_\perp=0} = \text{Im}(E) |\psi_0|^2 / \kappa_r$. Eq. (10) is therefore equivalent, for arbitrary finite polynomial order in $k_\perp$, to the single real requirement that the surface-state energy be real, $\text{Im}(E) = 0$ – precisely the self-adjointness statement that a genuine surface-bound state must have a real eigenvalue – which reduces to the familiar single-field current condition of Eq. (12), $\psi_0^\dagger j_{r_\perp} \psi_0|_{r_\perp=0} = 0$, whenever H_N is at most linear in $k_\perp$, as for the Weyl Hamiltonian.”

We break the underlying derivation up into the following steps, through the addition of an appendix:

- Step 1 (set-up):

We write the bulk Hamiltonian as a matrix polynomial in the perpendicular momentum,

$$H_N(k_\perp, \mathbf{k}_\parallel) = \sum_{n \geq 0} C_n(\mathbf{k}_\parallel) k_\perp^n.$$

Hermiticity of H_N for every real $k_\perp$ requires each C_n to be a Hermitian matrix. Throughout the derivation, $k_\perp$ should be understood as the differential operator, $-i \partial_{r_\perp}$.

- Step 2 (a Lagrange/Wronskian identity for a single power of $k_\perp$):

By induction on the degree n , one establishes the operator identity

$$\psi^\dagger(r_\perp) [k_\perp^n \phi(r_\perp)] - [k_\perp^n \psi(r_\perp)]^\dagger \phi(r_\perp) = -i \partial_{r_\perp} \sum_{m=0}^{n-1} [k_\perp^m \psi(r_\perp)]^\dagger [k_\perp^{n-1-m} \phi(r_\perp)]. \quad (1)$$

This identity generalises the familiar integration-by-parts formula used for the Weyl-node Hamiltonian. The base case $n = 1$ is immediate, $\psi^\dagger(k_\perp \phi) - (k_\perp \psi)^\dagger \phi = -i \partial_{r_\perp} (\psi^\dagger \phi)$, and is exactly the manipulation used implicitly for the linear-in- $k_\perp$ Weyl Hamiltonian. Assuming Eq. (1) holds up to degree $(n - 1)$, writing $k_\perp^n \phi = k_\perp (k_\perp^{n-1} \phi)$, and adding and subtracting $(k_\perp^{n-1} \psi)^\dagger (k_\perp \phi)$ reduces the degree- n identity to the base case (applied to the pair, ψ and $k_\perp^{n-1} \phi$) plus the degree- $(n - 1)$ identity (applied to the pair $k_\perp \psi$ and ϕ) — this completes the induction and fixes the coefficients on the right-hand side of Eq. (1) uniquely.

- Step 3 (boundary functional):

Multiplying Eq. (1) by the constant (i.e., $r_\perp$ -independent) Hermitian matrix C_n , summing over n , and integrating over the half-space gives

$$\psi^\dagger(r_\perp) H_{r_\perp} \phi(r_\perp) - (H_{r_\perp} \psi(r_\perp))^\dagger \phi(r_\perp) = -i \partial_{r_\perp} \Xi[\psi, \phi](r_\perp), \quad \Xi[\psi, \phi] \equiv \sum_n \sum_{m=0}^{n-1} (k_\perp^m \psi)^\dagger C_n (k_\perp^{n-1-m} \phi). \quad (2)$$

Integrating over $r_\perp \in [0, \infty)$, the contribution at $r_\perp \rightarrow \infty$ vanishes because $\psi, \phi \propto e^{-\kappa r_\perp}$ (with $\text{Re}(\kappa) > 0$) decay exponentially together with all their derivatives, so that Eq. (10) becomes simply $\Xi[\psi, \phi](0) = 0$.

- Step 4 (physical boundary state):

It remains to evaluate $\Xi[\psi, \phi](0)$ on the actual candidate surface state, $\psi(r_\perp) = \phi(r_\perp) = \psi_0 e^{-\kappa r_\perp}$. Writing $\lambda \equiv i\kappa$, so that $k_\perp^m \psi_0 = \lambda^m \psi_0$ for every m (since $-i \partial_{r_\perp} e^{-\kappa r_\perp} = i\kappa e^{-\kappa r_\perp}$). Here one must be careful: normalisability of the surface state fixes only $\kappa_r \equiv \text{Re}(\kappa) > 0$, while $\kappa_i \equiv \text{Im}(\kappa)$ is, in general, an independent real unknown solved for jointly with E [cf. the parametrisation below Eq. (11)]. Consequently $\text{Im}(\lambda) = \kappa_r > 0$ strictly, so λ is never real, and $(k_\perp^m \psi_0)^\dagger = \bar{\lambda}^m \psi_0^\dagger$, with $\bar{\lambda} \neq \lambda$ in general. The m -sum defining Ξ must therefore be summed as the (generally non-degenerate) finite geometric series it is,

$$\sum_{m=0}^{n-1} \bar{\lambda}^m \lambda^{n-1-m} = \frac{\lambda^n - \bar{\lambda}^n}{\lambda - \bar{\lambda}}, \quad \lambda \neq \bar{\lambda}, \quad (3)$$

rather than collapsing termwise to $n \lambda^{n-1}$. This gives

$$\Xi[\psi_0, \psi_0](0) = \sum_n \psi_0^\dagger C_n \psi_0 \frac{\lambda^n - \bar{\lambda}^n}{\lambda - \bar{\lambda}} = \frac{\psi_0^\dagger H_N(\lambda) \psi_0 - \psi_0^\dagger H_N(\bar{\lambda}) \psi_0}{\lambda - \bar{\lambda}}, \quad (4)$$

where $H_N(\lambda) \equiv \sum_n C_n \lambda^n$ denotes the matrix polynomial $H_N(k_\perp, \mathbf{k}_\parallel)$, with $k_\perp \rightarrow \lambda$ (a c-number substitution, not an operator one). Now, ψ_0 solves the eigenvalue equation of Eq. (13) specialised to this ansatz, $H_N(\lambda) \psi_0 = E \psi_0$ — hence, $\psi_0^\dagger H_N(\lambda) \psi_0 = E |\psi_0|^2$. Since $C_n^\dagger = C_n$, we have $H_N(\bar{\lambda}) = H_N(\lambda)^\dagger$, leading to $\psi_0^\dagger H_N(\bar{\lambda}) \psi_0 = [H_N(\lambda) \psi_0]^\dagger \psi_0 = E^* |\psi_0|^2$. Using $\lambda - \bar{\lambda} = 2i \text{Im}(\lambda) = 2i \kappa_r$, we obtain the exact identity,

$$\Xi[\psi_0, \psi_0]|_{r_\perp=0} = \frac{(E - E^*) |\psi_0|^2}{2i \kappa_r} = \frac{\text{Im}(E)}{\kappa_r} |\psi_0|^2. \quad (5)$$

Eq. (10) (i.e., $\Xi(0) = 0$, from Step 3) is therefore equivalent, order by order, and irrespective of the polynomial degree of H_N in $k_\perp$, to the single real condition, $\text{Im}(E) = 0$: the Hermiticity/hard-wall requirement is precisely the statement that a genuine surface-bound state must carry a real energy eigenvalue, as any state in the domain of a self-adjoint operator must. This is the general form of the boundary condition within the class of polynomial continuum Hamiltonians considered here, and is the condition actually enforced [as one real equation among the $(2N+1)$ counted below Eq. (13)]. It reduces to the familiar statement of Eq. (12), $\psi_0^\dagger j_{r_\perp} \psi_0|_{r_\perp=0} = 0$, precisely when H_N is at most linear in $k_\perp$ [i.e., $H_N = C_1 k_\perp + C_0(\mathbf{k}_\parallel)$, as for the untilted Weyl Hamiltonian]: there, $j_{r_\perp} = \partial_{k_\perp} H_N = C_1$ is κ -independent and Hermitian, the m -sum contains the single term $m = 0$ (so no $\lambda, \bar{\lambda}$ dependence enters at all), and $\Xi = \psi_0^\dagger C_1 \psi_0$ is already manifestly real — consistent with, and equal to, $\text{Im}(E) |\psi_0|^2 / \kappa_r$ above.

The same result also follows directly from the underlying physical continuity equation, independently of the geometric-sum bookkeeping above. Since $\psi(r_\perp)$ solves $H_{r_\perp} \psi(r_\perp) = E \psi(r_\perp)$ pointwise, we have $\psi^\dagger(r_\perp) H_{r_\perp} \psi(r_\perp) = E |\psi(r_\perp)|^2$ and $(H_{r_\perp} \psi(r_\perp))^\dagger \psi(r_\perp) = E^* |\psi(r_\perp)|^2$, so that the Step 3 identity, specialised to $\phi = \psi$, reads $(E - E^*) |\psi(r_\perp)|^2 = -i \partial_{r_\perp} \Xi[\psi, \psi](r_\perp)$, i.e., $\partial_{r_\perp} \Xi[\psi, \psi](r_\perp) = -2 \text{Im}(E) |\psi(r_\perp)|^2$. Integrating from $r_\perp = 0$ to ∞ , using $\Xi[\psi, \psi](\infty) = 0$ by decay and $\int_0^\infty |\psi_0|^2 e^{-2\kappa_r r_\perp} dr_\perp = |\psi_0|^2 / (2\kappa_r)$, reproduces $\Xi[\psi_0, \psi_0](0) = \text{Im}(E) |\psi_0|^2 / \kappa_r$ exactly, for any polynomial degree of H_N . We emphasise that this is *not* the same statement as $\text{Re}[\psi_0^\dagger j_{r_\perp} \psi_0] = 0$: the operator $j_{r_\perp} = \partial_{k_\perp} H_N$ is Hermitian only when evaluated at real $k_\perp$, whereas at the boundary it is evaluated at the complex value $k_\perp \rightarrow \lambda$, so that $\psi_0^\dagger j_{r_\perp} \psi_0|_{r_\perp=0}$ is an analytic continuation rather than a genuine Hermitian expectation value, and need not coincide with the manifestly real $\Xi[\psi_0, \psi_0](0)$ beyond quadratic order in $k_\perp$, as we illustrate explicitly below.

As an explicit illustration beyond the Weyl case, consider the double-Weyl Hamiltonian of Eq. (26), for which C_0, C_1, C_2 are all nonzero, so that H_N is genuinely quadratic in $k_\perp = k_x$. For any polynomial containing only powers up to $n = 2$, the geometric sum of Step 4 simplifies further: since the $n = 2$ factor obeys $(\lambda^2 - \bar{\lambda}^2)/(\lambda - \bar{\lambda}) = \lambda + \bar{\lambda} = 2 \text{Re}(\lambda)$ for any complex λ , the $n = 2$ contribution to $\Xi[\psi_0, \psi_0](0)$ reduces to $2 \text{Re}(\lambda) \psi_0^\dagger C_2 \psi_0$. Hence, it coincides exactly with the real part of the corresponding term of $\psi_0^\dagger j_{r_\perp} \psi_0$. Combined with the $n \leq 1$ terms, which are already manifestly real, this gives $\Xi[\psi_0, \psi_0](0) = \text{Re}[\psi_0^\dagger j_{r_\perp} \psi_0]|_{r_\perp=0}$, whenever H_N is at most quadratic in $k_\perp$ — thus, $\text{Im}(E) = 0$ is equivalent, for

such Hamiltonians, to the real part of the current condition of Eq. (12), even though κ itself remains genuinely complex. Using $\psi_0^T = [1 \quad a_1 + i b_1]$ and $j_x = \partial_{k_x} H_{dW} = 2 k_x \sigma_x + 2 k_y \sigma_y$ (with k_0 set to 1), one finds $\psi_0^\dagger \sigma_x \psi_0 = 2 a_1$ and $\psi_0^\dagger \sigma_y \psi_0 = 2 b_1$, so that $\psi_0^\dagger j_x \psi_0|_{x=0} = 2 i \kappa (2 a_1) + 2 k_y (2 b_1) = (4 b_1 k_y - 4 a_1 \kappa_i) + 4 i a_1 \kappa_r$, whose real part vanishing gives exactly $\kappa_i = (b_1/a_1) k_y$ [reproducing Eq. (29) verbatim]. This confirms explicitly, for the case in the paper where H_N is quadratic, that the general condition $\text{Im}(E) = 0$ derived above reduces to the boundary condition already used in Sec. III.E. For Hamiltonians containing $k_\perp^n$ with $n \geq 3$ (e.g., the triple- and quadruple-Weyl models), this intermediate identification with $\text{Re}[\psi_0^\dagger j_{r_\perp} \psi_0]$ need not hold term by term; there, $\text{Im}(E) = 0$ itself, established generally in Step 4 above, remains the correct boundary condition.

In finalising this derivation, we have made two further small refinements for clarity and rigour. First, in the double-Weyl illustration above, the $n = 2$ reduction is now justified directly from the Step-4 geometric-sum factor, $(\lambda^2 - \bar{\lambda}^2)/(\lambda - \bar{\lambda}) = \lambda + \bar{\lambda} = 2 \text{Re}(\lambda)$, rather than via an unrelated identity for $\text{Im}(\lambda^2)$ that we had cited previously; the final result is unchanged. Second, since the triple-Weyl Hamiltonian of Sec. III.G is cubic in $k_\perp$, we have added a sentence there clarifying that Eq. (12) is invoked in the general (Appendix) sense, $\Xi[\psi_0, \psi_0](0) = 0$, rather than as a literal evaluation of the current operator at the complex boundary value $k_\perp \rightarrow \lambda$ – consistent with the point just established that the two need not coincide beyond quadratic order. This avoids any impression that Sec. III.G follows from a naive reading of Eq. (12), which, as shown above, would not in general reproduce the correct boundary condition for a cubic Hamiltonian.

3. Besides, the types of the Weyl nodes should be classified according to the values of the topological invariant N_3 responsible for the stability of Fermi points (the invariant composed of the Green functions). Next, the values of this N_3 should be calculated for all examples presented in Sect. III.

We have added the following discussions:

“For a non-interacting Hamiltonian in the vicinity of a nodal point, the relevant Matsubara Green’s function is $G(i\omega, \mathbf{k}) = [i\omega - H_N(\mathbf{k})]^{-1}$, and the invariant protecting the point node against gap opening under continuous deformations is Volovik’s N_3 [1], defined as the degree of the map from a three-sphere S^3 surrounding the singularity in $(\omega, \mathbf{k})$ -space into the space of Green’s functions,

$$N_3 = \frac{1}{24\pi^2} \epsilon^{\mu\nu\rho\sigma} \text{tr} \oint_{S^3} dS^\mu G \partial_\nu G^{-1} G \partial_\rho G^{-1} G \partial_\sigma G^{-1}. \quad (6)$$

The integration is performed over S^3 . For the effective low-energy Hamiltonians considered throughout the manuscript, N_3 reduces identically to the integral of the Berry curvature (BC) flux over any Gaussian surface (e.g., S^2) enclosing the node, where the BC of the s -th band belonging to the eigenvalue χ of the parent Hamiltonian is

$$\Omega_s(\mathbf{k}) = i [\nabla_{\mathbf{k}} \psi_s(\mathbf{k})]^\dagger \times [\nabla_{\mathbf{k}} \psi_s(\mathbf{k})], \quad (7)$$

with $\{\psi_s(\mathbf{k})\}$ being the set of normalised eigenvectors of the parent Hamiltonian. We can now use this to compute N_3 directly, as the flux of $\Omega_s(\mathbf{k})$ through any closed surface enclosing the node. Equivalently, this is the BC-monopole charge or Chern number that we already invoke in Sec. III to count the number of Fermi arcs, and has already been computed over and over in the literature (e.g., Refs. [2–11]). The reduction from the frequency-momentum S^3 -integral to the equal-time S^2 -integral is standard for gapless free-fermion Green’s functions and can be computed using Eq. (7). Having made this identification explicit, we now tabulate N_3 for every variety of node discussed in this paper:

Type of node	Band-multiplicity (N)	$ N_3 = \mathcal{C} $
Untilted Weyl node	2	± 1
Tilted Weyl node	2	± 1
Triple-point node	3	± 2
RSW node	4	± 4
Double-Weyl node	2	± 2
Triple-Weyl node	2	± 3
Berry-dipole (two-band)	2	0
Berry-dipole (three-band)	3	0
Quadruple-Weyl node	2	± 4

The two BC-dipole cases carry vanishing net monopole charge, $N_3 = 0$, by construction, since an ideal BC-dipole has equal and opposite monopole charge on its two poles; nonetheless, as we discuss in the manuscript, they still support Fermi arcs, since our boundary-condition construction is sensitive to more than the net charge alone.”

4. I suspect that these values should coincide for some of the cases. If so, what is the difference from them (from the point of view of topology)?

The referee’s suspicion is correct, and we have added a discussion of this point to the manuscript:

“From our analysis, we observe the following: While N_3 (and, hence, $\mathcal{C}$) is the topological invariant governing the stability of the point node itself, the existence and multiplicity of the associated Fermi arcs are more directly controlled by the

Chern numbers $\mathcal{C}(k_z)$ of the family of 2d Hamiltonians obtained by slicing the BZ at fixed values of one momentum component parallel to the surface (say, k_z). For the ideal BC-dipole nodes, $N_3 = 0$, while individual slices retain $\mathcal{C}(k_z) = \pm 1$ over an extended range of k_z , which is what permits Fermi arcs to survive despite the vanishing point-node charge. Nodes sharing the same value of N_3 are topologically equivalent in the sense of carrying the same number of protected arc branches, but generically differ in band-multiplicity and in the momentum-power/anisotropy of the dispersion, both of which are non-universal and set the detailed curvature of the arcs rather than their protected count. We provide below a comparison on a case by case:

- *Untilted versus tilted Weyl node:* The tilt term $\eta k_x \mathbb{I}_{2 \times 2}$ existing in Eq. (17) is proportional to the identity matrix and therefore does not alter the eigenvectors of $H_W(\mathbf{k})$ at all — thus the associated BC and, hence, $N_3 = \pm 1$ is completely unchanged. The visibly different curvature of the resulting Fermi arcs in the two cases is thus not a topological effect at all, but simply reflects the anisotropic reshaping of the bulk Fermi surface (and of its projection) by the tilt.
- *Triple-point node versus double-Weyl node:* Both carry $N_3 = \pm 2$ and, consistently, both source exactly two Fermi arcs per node. However, the triple-point node is a threefold-degenerate isotropic crossing, while the double-Weyl node is a twofold-degenerate crossing with an anisotropic dispersion. The value of N_3 , being a total flux (i.e., an integral quantity), is insensitive to this difference in band-multiplicity and angular/radial distribution of the underlying BC — it only counts the net number of protected chiral branches guaranteed by bulk-boundary correspondence, and not their microscopic origin. This is precisely why the two arc families look qualitatively different (smoothly curving for the triple point, sharply bent for the double-Weyl node) — the detailed curvature of an individual arc is controlled by non-topological, continuously tunable data [e.g., the local dispersion and the boundary-condition parameter(s)], rather than by N_3 itself.
- *RSW node versus quadruple-Weyl node:* Both carry $N_3 = \pm 4$ and both source four Fermi arcs, yet the RSW node is a fourfold-degenerate fully-isotropic linear crossing, whereas the quadruple-Weyl node is twofold-degenerate with non-linear anisotropic dispersion. Correspondingly, the bulk FSs and, thus, the shapes of their projections onto the SBZ, are entirely different in the two cases. In the quadruple-Weyl case, the projected FS is not even a simple curve of the form $k = \mu$, but instead is a more intricate closed contour. Once again, N_3 dictates only the topologically-protected count of arcs (four, in both cases), while their detailed geometry is fixed by the non-universal bandstructure.
- *BC-dipole node:* Here, $N_3 = 0$ for the node as a whole (since it is an ideal dipole) and yet Fermi arcs are found to exist. This somewhat counter-intuitive result is not in tension with the vanishing of N_3 : while the point-node invariant N_3 only characterises the net flux through an S^2 surrounding the full 3d singularity, the arcs we find are protected by a finer-grained momentum-resolved invariant [namely, the Chern number $\mathcal{C}(k_z)$ of each 2d constant- k_z slice of the BZ, viewed as an independent 2d Hamiltonian $H_N(k_x, k_y; k_z)$]. For a dipole, $\mathcal{C}(k_z)$ takes values of ± 1 on pairwise slices. This makes $N_3 = 0$, even though $\mathcal{C}(k_z) \neq 0$ for a continuous range of individual slices — it is precisely this slice-resolved topology that our formalism accesses.

To summarise, nodes sharing the same N_3 are, from the point of view of the bulk topological classification alone, indeed indistinguishable, since N_3 is a coarse integer invariant that only counts the total winding of the Green's function (equivalently, the net Berry flux) around the node, and says nothing about how that total charge is distributed among the internal (pseudospin/band) degrees of freedom, or (for the dipole cases) about how it is distributed in k_z . Our boundary-condition formalism, however, is sensitive to more than N_3 alone: it depends on the full polynomial structure of H_N , in particular on the band multiplicity N and on the power(s) of momentum through which the monopole charge is realised, and, for the dipole cases, on the slice-resolved $\mathcal{C}(k_z)$ itself. These are genuinely different, topologically non-equivalent routes to the same net charge — they cannot be continuously deformed into one another without closing additional gaps or changing the band multiplicity. This is exactly why, despite sharing the same N_3 , the various nodes enter our generic construction with different N , different M (the highest power of $k_\perp$ retained), and different matrix coefficients. Consequently, they yield Fermi arcs of visibly-different shapes, curvature, and parameter dependence, even though the number of arcs terminating at each FS-projection (fixed by $|N_3|$, where non-zero) is the same.”

5. Even more important is to compare the obtained patterns of the Fermi arcs to the forms of the Fermi arcs observed by ARPES for various topological semimetals. Indeed often these experimental patterns look very interesting and the question arises, why these forms are so bizarre. If the author's finding will help us to answer this question, the paper might be interesting for the audience.

We have added the following paragraph to the last section:

“It is instructive to compare the arc families derived above with Fermi arcs resolved by ARPES. The number of arc branches emanating from a given nodal projection is fixed by the topological invariant N_3 of the node (equivalently, by our tabulated, slice-resolved Chern numbers). ARPES measurements on WSMs of the TaAs family reveal short, simply-curved Fermi arcs consistent with $N_3 = \pm 1$. Chiral crystals such as CoSi, RhSi, and PtGa, on the other hand, host multifold nodal degeneracies (or higher-charge nodes), exemplified by our RSW and pseudospin-1 models. These display long Fermi arcs that traverse an appreciable fraction of the SBZ and connect nodes of opposite chirality located at different high-symmetry points [12–14], consistent with $|N_3| = 4$ obtained here. Both features are captured naturally by our results: the number of arc branches sourced at a given projected FS is fixed by N_3 , exactly as observed — single

arcs for TaAs-like materials, multiple/longer arcs for the higher-charge multifold semimetals. This protected count is entirely fixed by bulk topology and is insensitive to the boundary-condition parameter. By contrast, the pronounced curvature, kinking, and sometimes counter-intuitive (“bizarre”) shape of individual arcs seen within a single ARPES data set, even for nominally the same bulk charge, is controlled by our boundary-condition parameters in the notation of Sec. III. These parameters are fixed by the microscopic details of the surface termination and, therefore, expected to vary between samples, cleaves, or materials sharing the same bulk topology. In other words, the qualitative topologically-protected features of Fermi arcs (their number and connectivity to bulk Fermi-surface projections) are governed by N_3 . Their strongly-curved sample-dependent appearance in ARPES reflects the continuous family of admissible boundary conditions instead, which are a direct symmetry-allowed consequence of surface reconstruction/termination chemistry. Our generic construction shows explicitly how a topologically fixed number of arcs can nonetheless be continuously deformed in shape by these parameters. This offers a concrete mechanism for the diversity of arc geometries reported experimentally for materials sharing the same bulk nodal charge, parametrising this correspondence explicitly and analytically for the first time across this range of nodal types.”

II. RESPONSE TO QUESTIONS/COMMENTS OF REFEREE #2

1. The manuscript presents an analytical study of Fermi arcs in generic nodal semimetals and discusses a unified formulation for surface states in systems with different monopole or dipole structures. The topic is interesting and potentially useful for understanding boundary states in topological semimetals. However, I have an important concern regarding the formulation of the boundary conditions in Sec. II, which should be clarified before publication.

We thank the referee for the positive assessment of the topic and for the careful reading of Sec. II. We address the specific concerns raised, point by point, below.

2. In Eq. (2), the author introduces two fields, $\psi(x)$ and $\phi(x)$. However, the physical meaning and role of these two fields are not explained clearly. In particular, it is not obvious why the boundary condition is formulated in terms of both ψ and ϕ , while the later hard-wall condition is eventually imposed only on ϕ (or effectively on a single wavefunction). The authors should explain more carefully the mathematical and physical origin of this procedure.

On the semi-infinite domain $x \in [0, \infty)$, the boundary Hamiltonian H_x is a differential operator, with $H_x = H(k_x \rightarrow -i \partial_x, k_y, k_z)$. It is only formally symmetric on an arbitrary space of decaying functions on $[0, \infty)$. Whether it is genuinely self-adjoint — and hence generates well-defined dynamics with a real, discrete spectrum of surface bands — depends on which boundary data are allowed at $x = 0$. This is exactly the textbook problem of fixing self-adjoint extensions of a differential operator on a half-line via von Neumann’s deficiency-index construction, and has been discussed explicitly for Weyl-semimetal boundaries in Ref. [15]. The standard definition of a symmetric operator requires $\int_0^\infty dx \psi^\dagger(x) H_x \phi(x) = \int_0^\infty dx [H_x \psi(x)]^\dagger \phi(x)$ to hold for every pair ψ, ϕ drawn from the domain of decaying functions on $[0, \infty)$ — not for a single, specific state. This is precisely why Eq. (2) is written with two independent, generic functions $\psi(x)$ and $\phi(x)$. They are simply two independent, otherwise arbitrary, members of this function space, introduced solely as test functions in the conventional sense of self-adjoint-extension theory. They carry no separate physical meaning of their own; they are not to be thought of as “incoming” and “outgoing”, or “bra” and “ket”, states in any dynamical sense. Demanding this identity for arbitrary, independent ψ, ϕ is precisely what forces the bilinear boundary term $(\psi^\dagger j_x \phi)|_{x=0}$ (with $j_x = \partial_{k_x} H|_{k_x \rightarrow -i \partial_x}$) to vanish identically as a bilinear form. This is exactly the origin of the hard-wall-type boundary condition, and in turn fixes the linear space of admissible spinor components at $x = 0$, i.e., the boundary-condition matrix M of Eq. (7). Crucially, this matrix M , and the resulting energy-independent boundary condition $\Psi_0 = M \Psi_0$, is a property of the operator’s domain alone. It must be satisfied by every admissible boundary state, independent of that state’s energy or momentum. Fixing a boundary condition valid for the entire domain, rather than for one accidental solution, is precisely a statement about a bilinear form vanishing on the whole function space, which is only meaningful when tested against independent pairs. This is why the derivation genuinely requires two independent functions ψ, ϕ at this stage. Once M (or, equivalently, the general current condition) has been fixed once and for all in this way, any candidate surface-bound state ψ must itself satisfy this very same boundary relation, in order to be an admissible member of the self-adjoint domain. Here ψ is obtained separately by solving the eigenvalue equation $H_{r_\perp} \psi = E \psi$ [Eq. (13)] together with the exponential-decay ansatz $\psi(x) = \psi_0 e^{-\kappa x}$. This is the statement $\psi_0^\dagger j_x \psi_0|_{x=0} = 0$ used throughout Sec. III: it is simply the general two-field boundary condition above, specialised to the case $\phi = \psi$, i.e., applied to the specific solution one has just constructed, rather than a new or weaker condition. Thus ϕ is not an independent auxiliary field surviving into the final answer; it is the second, independent test function needed once, to determine the admissible class of boundary matrices M , and it collapses onto the physical surface state once we ask for the condition obeyed by an actual candidate boundary eigenstate.

We have added the following discussions:

“To avoid any confusion, we emphasise that $\psi(x)$ and $\phi(x)$ appearing in Eq. (2) denote two independent and arbitrary decaying functions on $[0, \infty)$. They are introduced solely as test functions in the conventional sense of self-adjoint-extension theory, as previously employed for WSM boundaries in Ref. [15]. The operator H_x itself remains Hermitian for all real momenta throughout and no non-Hermitian structure is implied. Demanding the Hermiticity identity of Eq. (2) for arbitrary independent $\{\psi, \phi\}$ is what fixes the admissible, energy-independent boundary condition [see also

Eq. (7) below]. This defines a genuinely self-adjoint domain for H_x on the half-space ($x < 0$), analogous to fixing self-adjoint extensions of a differential operator via von Neumann's deficiency-index construction. Once this domain is fixed, any candidate surface-bound eigenstate must satisfy the same relation with ψ set equal to itself, in order to belong to it. The resulting single-field current condition, $\psi_0^\dagger j_x \psi_0|_{x=0} = 0$, used throughout this paper, is thus not a distinct or weaker requirement. It is precisely this general boundary condition evaluated on the specific physical solution at hand – the standard hard-wall statement that the constructed surface state carries no probability current through the boundary.”

3. Related to this, it is also unclear why the boundary condition only constrains the current operator through one state, rather than symmetrically involving both ψ and ϕ . Is this related to a non-Hermitian-type formulation, self-adjoint extension theory, or simply a conventional Hermiticity requirement for differential operators with open boundaries?

This follows directly from the two-step logic described above: The two-field version is what one uses once, to determine the admissible class of boundary matrices M [or, equivalently, the general structure of Eq. (12)]; the one-field version used repeatedly in Sec. III is simply this same condition evaluated on the actual physical solution. The boundary term derived from Eq. (2) is, in general, a bilinear (sesquilinear) form in the two independent functions ψ and ϕ , and would indeed involve both symmetrically if we were testing the Hermiticity of H_x in an abstract mathematical way (over the applicable domain). However, the quantity of physical interest at the second stage is not this abstract bilinear form, but the specific condition obeyed by an actual candidate surface state. Since only one physical wavefunction, $\psi(x) = \psi_0 e^{-\kappa x}$, is being sought at that stage of the calculation, the natural and unique choice is to evaluate the boundary term on this single state by setting $\phi \rightarrow \psi$. The resulting current condition, $\psi^\dagger j_x \psi|_{x=0} = 0$, therefore constrains one state because there is only one physical state being tested at that point; physically, this is exactly the hard-wall statement that the state must carry no probability current through the boundary at $x = 0$ – a necessary condition for a bona fide bound (evanescent) surface state, entirely analogous to demanding $j(x) \propto \text{Im}[\psi^*(x) \partial_x \psi(x)] = 0$ at the wall of an ordinary infinite square well, generalised here to a matrix current appropriate for the multiband, higher-momentum-power Hamiltonians considered in this paper.

We confirm explicitly that the entire construction is a conventional Hermiticity (self-adjointness) requirement for a differential operator on the half-line $x \in [0, \infty)$. It is in no way related to a non-Hermitian formulation; H_x itself, as a matrix function of real momentum, is Hermitian throughout. What is at stake is the choice of *domain*: fixing which boundary data are allowed at $x = 0$, so that H_x is not merely formally symmetric but genuinely self-adjoint. This is exactly the textbook problem of fixing self-adjoint extensions of the momentum or Laplacian operator on a half-line or an interval (von Neumann's deficiency-index construction), and has been discussed explicitly for Weyl-semimetal boundaries in Ref. [15]. This is the underlying form needed to isolate the surface term and, thereby, derive the physical no-current-leakage condition that any genuine normalisable surface state must satisfy. This is concretely realised, for the isotropic Weyl node, as Witten's M -matrix parametrisation of Eq. (7). The bulk Hamiltonian itself remains strictly Hermitian throughout; only its restriction to a domain with a boundary requires this additional care.

4. At present, the logic is difficult to follow. I encourage the author to provide a clearer derivation and physical discussion of Eq. (2), including the meaning of the auxiliary field $\phi(x)$ and the precise justification of the adopted boundary conditions.

The operator H_x itself remains Hermitian for all real momenta throughout, and no non-Hermitian structure is implied, as discussed in our response to point 2 of Referee #1 above (conforming to which details of the derivation has been added as an appendix). We have revised the text to state this explicitly:

“To avoid any confusion, we emphasise that $\psi(x)$ and $\phi(x)$ appearing in Eq. (2) denote two independent and arbitrary decaying functions on $[0, \infty)$. They are introduced solely as test functions in the conventional sense of self-adjoint-extension theory, as previously employed for WSM boundaries in Ref. [15]. The operator H_x itself remains Hermitian for all real momenta throughout and no non-Hermitian structure is implied. Demanding the Hermiticity identity of Eq. (2) for arbitrary independent $\{\psi, \phi\}$ is what fixes the admissible, energy-independent boundary condition [see also Eq. (7) below]. This defines a genuinely self-adjoint domain for H_x on the half-space ($x < 0$), analogous to fixing self-adjoint extensions of a differential operator via von Neumann's deficiency-index construction. Once this domain is fixed, any candidate surface-bound eigenstate must satisfy the same relation with ψ set equal to itself, in order to belong to it. The resulting single-field current condition, $\psi_0^\dagger j_x \psi_0|_{x=0} = 0$, used throughout this paper, is thus not a distinct or weaker requirement. It is precisely this general boundary condition evaluated on the specific physical solution at hand – the standard hard-wall statement that the constructed surface state carries no probability current through the boundary.”

III. LIST OF CHANGES

We have marked all additions in the manuscript with the colour red.

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